

Skin Severity Assessment — Parameters & Decision Methodology

Internal reference · classical CV parameter set, Weighted
Severity Index, and remedy matrix

CONFIRMED & IMPLEMENTED

SCOPE — ACNE · PIGMENTATION · WRINKLE

VERIFIED AGAINST FULL 27-CASE DECISION MATRIX

01 Overview

Each analysis reads a set of **measurable parameters** per condition from the captured photo, bands each one Mild / Moderate / Severe against a fixed clinical threshold, then combines the three bands into one severity call using the **Weighted Severity Index (WSI)**. That severity — Mild, Moderate, or Severe — is the sole lookup key into the remedy matrix. This document is the confirmed record of what's measured, how it's weighted, and how the decision and remedy follow from it.

02 Acne

Three parameters drive the decision; a fourth (lesion type) acts as a clinical safety override, not a voting input.

PARAMETER	MEASURES	MILD	MODERATE	SEVERE
Lesion count lesion_count_per_cm ² — primary, weight 0.5	Lesions per unit area	< 6	6 – 15	> 15
Inflammatory % inflammatory_pct — density proxy, weight 0.25	Share of lesions that are pustules/nodules	< 20%	20 – 50%	> 50%
Redness Index redness_index — tertiary, weight 0.25	Erythema Index, EI = (R-G)/(R+G+B)	< 15%	15 – 35%	> 35%
Lesion type comedones / papules_pustules / nodules_cysts	Morphology classification	Not banded — see override below		

SAFETY OVERRIDE

Any lesion classified as **nodules_cysts** forces the decision to **Severe**, regardless of what count, inflammatory %, or redness independently indicate.

03 Pigmentation

All three parameters vote; two combinations carry a hard override to Severe.

PARAMETER	MEASURES	MILD	MODERATE	SEVERE
% pigmented area pigmented_area_pct — primary, weight 0.5	Pigmented pixels as % of ROI	< 10%	10 – 30%	> 30%
ΔGray (intensity) delta_gray — secondary, weight 0.25	Contrast vs. adjacent normal skin	< 20	20 – 50	> 50
Melanin Index melanin_index — tertiary, weight 0.25	MI = 100 · log ₁₀ (255 / R)	< 50	50 – 100	> 100

SAFETY OVERRIDES

Pigmented area > 50% → Severe. Melanin Index > 150 and ΔGray > 80 → Severe (very dark, high-contrast pigmentation).

04 **Wrinkle**

Same three-parameter vote; no lesion-type equivalent exists, so there's no morphology-based hard override here.

PARAMETER	MEASURES	MILD	MODERATE	SEVERE
Wrinkle count wrinkle_count_per_cm² — primary, weight 0.5	Skeletonized lines per unit area	< 5	5 – 10	> 10
Length avg_length_mm — secondary, weight 0.25	Mean skeleton line length	< 20 mm	20 – 40 mm	> 40 mm
Depth (ΔGray) depth_delta_gray — tertiary, weight 0.25	Trough-vs-skin intensity contrast	< 20	20 – 50	> 50

WRINKLE-SPECIFIC CAP

Count Mild + Length Severe + Depth Severe caps at **Moderate** rather than escalating to Severe — two lower-weight severses shouldn't outrank a mild primary reading.

05 **Weighted Severity Index**

The same formula and decision logic apply to all three conditions — only which measured field plays "primary / secondary / tertiary" changes.

$$\text{WSI} = 0.5 \times \text{Primary} + 0.25 \times \text{Secondary} + 0.25 \times \text{Tertiary}$$

Each parameter is first scored Mild = 0, Moderate = 0.5, Severe = 1, then combined by weight above.

PRIMARY



0.50

SECONDARY



0.25

TERTIARY



0.25

Acne

PRIMARY lesion_count_per_cm2

SECONDARY inflammatory_pct

TERTIARY redness_index

Pigmentation

PRIMARY pigmented_area_pct

SECONDARY delta_gray

TERTIARY melanin_index

Wrinkle

PRIMARY wrinkle_count_per_cm

SECONDARY avg_length_mm

TERTIARY depth_delta_gray

The **Decision** is a flat threshold on the WSI number itself: $< 0.33 \rightarrow \text{Mild}$, $0.33 - 0.66 \rightarrow \text{Moderate}$, $> 0.66 \rightarrow \text{Severe}$. On top of that sit a small number of clinical safety overrides that apply irrespective of WSI, plus an advisory flag whenever a lower-weighted parameter reads Severe on its own but gets outweighed in the overall number.

WSI $< 0.33 \rightarrow \text{Mild}$ · $0.33 \leq \text{WSI} \leq 0.66 \rightarrow \text{Moderate}$ · WSI $> 0.66 \rightarrow \text{Severe}$

APPLIES TO	TRIGGER	EFFECT
Acne only	Lesion type = nodules/cysts	Severe — hard override, irrespective of WSI
Pigmentation only	% area > 50%	Severe — hard override, irrespective of WSI
Pigmentation only	Melanin Index > 150 <i>and</i> Δ Gray > 80	Severe — hard override, irrespective of WSI
All three	Any parameter reads Severe on its own, but the WSI decision is Mild or Moderate	Decision unchanged — advisory flag names the elevated parameter(s)

06 Remedy matrix

The confirmed Decision is the only lookup key. Stored in MongoDB, editable by an admin at any time without a redeploy.

CONDITION	SEVERITY	RECOMMENDATION TYPE	EXAMPLES	DURATION
Acne	Mild	Home remedy	Cleanser, tea-tree oil, honey mask, low-sugar diet	2–4 wks
	Moderate	OTC cosmeceutical	Benzoyl peroxide 2.5–5% gel, niacinamide serum, salicylic acid 2% cleanser	4–8 wks
	Severe	Referral	Consult a dermatologist	–
Pigmentation	Mild	Home remedy	Aloe vera gel, green tea extract	2–4 wks
	Moderate	OTC cosmeceutical	Vitamin C serum, niacinamide 5–10%	4–8 wks
	Severe	Referral	Consult a dermatologist	–
Wrinkle	Mild	Home remedy	Aloe vera, honey, cucumber mask, sunscreen	2–4 wks
	Moderate	OTC cosmeceutical	Retinol 0.1–0.3%, hyaluronic acid serum	4–8 wks
	Severe	Referral	Consult a dermatologist	–

"For informational use only. Please consult a dermatologist for medical advice." — shown with every result.

07 End-to-end flow

01 Input Photo analyzed; three parameters measured per condition.	02 Decide WSI computed, threshold + safety overrides applied → Mild / Moderate / Severe.	03 Lookup Severity matched against the remedy matrix.	04 Output Remedy type, examples, and duration shown to the patient.
05 Track Before/after comparison once the duration elapses.			

Dermato – Skin Severity Assessment Methodology

Verified against the full decision matrix · all three conditions